

Question Paper

Exam Date & Time: 31-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

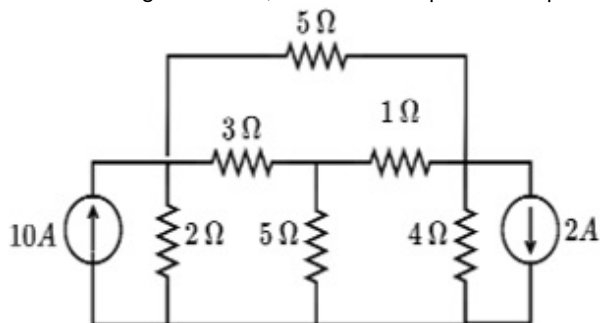
SUBJECT: ELE 1071/ELE_1071 BASIC ELECTRICAL TECHNOLOGY

Marks: 50

Duration: 180 mins.

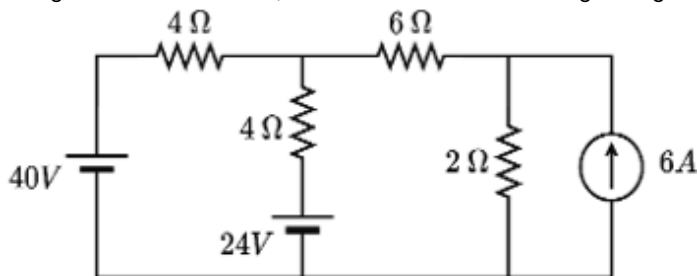
Answer all the questions.

- 1A) In the circuit given below, determine the power dissipated by the 3Ω resistor using mesh current analysis. (6)



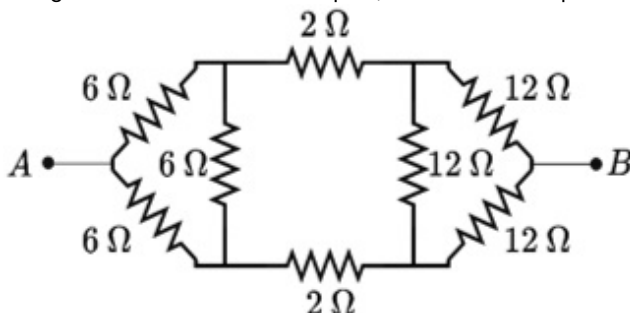
Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) Using Thevenin's theorem, determine the current flowing through the 6Ω resistor. (4)



Course Outcome: "CO1" / Cognitive Level: "Analysis"

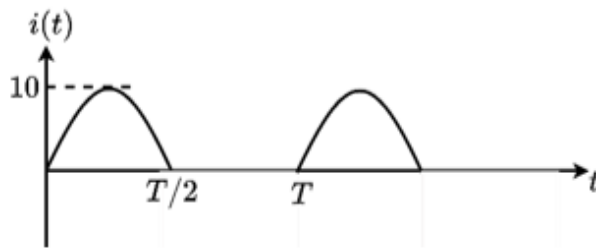
- 2A) Using network reduction techniques, determine the equivalent resistance between terminals A and B. (4)



Course Outcome: "CO1" / Cognitive Level: "Application"

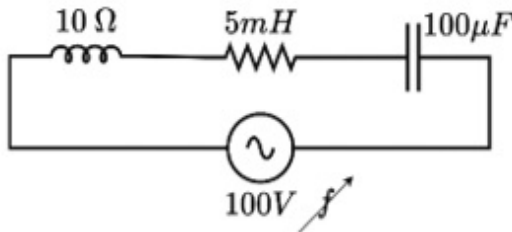
- 2B) Find the RMS value of the alternating current $i(t)$ corresponding to the waveform shown, (3)

considering a time period of $T = 20$ ms.



Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) A series RLC circuit, as shown below, is connected across a single-phase 100V, variable-frequency source. (3)



Determine:

- Resonant Frequency of the circuit.
- Quality Factor.
- Voltage across the capacitor at resonance.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Two loads with impedances $(24 + j32) \Omega$ and $(30 + j40) \Omega$ are connected in parallel across a single-phase 230 V, 50 Hz supply. Calculate, (6)
- Current through both loads
 - Total active power
 - Total reactive power
 - Capacitance of the capacitor to be connected across the parallel combination to raise the overall power factor of the circuit to 0.95 lagging.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3B) Two wattmeters are connected to measure the power in a 3-phase circuit, reading 5 kW and 1 kW, with the reading of the second wattmeter obtained after reversing the potential coil of the wattmeter. Calculate, (4)
- Total active power
 - Total reactive power
 - Power factor

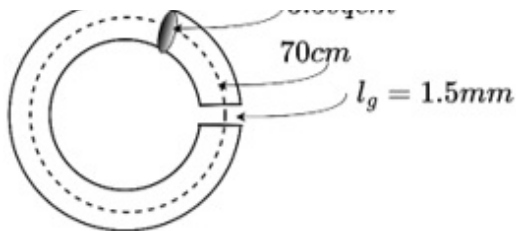
Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) A 3-phase, 3-wire, 415V, RYB supply is connected to a balanced star connected RL load of phase impedance $6 + j8 \Omega$. Taking V_{RN} as reference, calculate: (6)
- All line currents
 - Neutral shift voltage
 - Total active power
 - Draw the phasor diagram

Course Outcome: "CO3" / Cognitive Level: "Application"

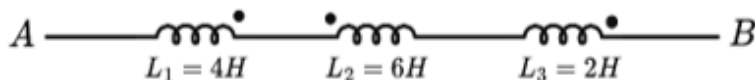
- 4B) A steel ring with a mean circumference of 70cm and a cross-section of 3.5 cm^2 has an air gap of 1.5mm in length. It is wound uniformly with a coil of 500 turns. Calculate the reluctance of the ring and the current required to produce a flux of 0.3mWb in the air gap. The relative permeability of the ferromagnetic material is 800. (4)

3.5 cm^2



Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) For the given magnetic coupling network, if the mutual inductances are $M_{12} = 2H$, $M_{23} = 1H$, and $M_{31} = 1H$, (3) calculate the coupling coefficients k_{12} , k_{23} , and k_{31} . Also, determine the equivalent inductance measured between terminals A and B.



Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) With a neat sketch, explain the structure of an electrical power system, highlighting the voltage levels at different stages. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) Draw the torque-slip characteristic of a three-phase induction motor and describe the different operating regions. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ELE_1071:BASIC ELECTRICAL TECHNOLOGY

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	12
3	3	CO 3	10
4	4	CO 4	7
5	5	CO 5	7
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ELE_1071:BASIC ELECTRICAL TECHNOLOGY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	7(14%)
3	Application	6	24(48%)
4	Analysis	4	19(38%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)